AYANDA PATRICK HERLIHY

apherlihy@gmail.com | Worcester, United Kingdom
Website

EDUCATION

Birmingham City University Awarded 21 August 2026
BSc (Hons) Computer Science with Artificial Intelligence
First Class Honours | Overall mark: 79.48% | Honours project: 79%
Dissertation: Geometric Deep Learning for Quantum Financial Market Dynamics
Supervisor: Dr Edlira Vakaj

Middlesex University / SAE Institute London Awarded 31 October 2018
Certificate of Higher Education in Audio Production

RESEARCH INTERESTS

Multi-agent systems; LLM agents; reinforcement learning; reasoning in language models, including chain-of-thought methods.

PUBLICATION

Jie JW Wu*, **Ayanda Patrick Herlihy***, Ahmad Saleem Mirza, Ali Afoud and Fatemeh Fard. GrowthHacker: Automated Off-Policy Evaluation Optimization Using Code-Modifying LLM Agents. *ACM Transactions on Software Engineering and Methodology (TOSEM), 2026.*
Accepted for publication. doi:10.1145/3815588 | Preprint
*Equal contribution.

RESEARCH EXPERIENCE

OPE-Bench August 2026–present
Research collaboration with Dr Jie Wu
- Developing OPE-Bench, a benchmark for evaluating LLM agents that modify off-policy evaluation code.
- Investigating the reliability of agents' claimed improvements and their performance on unseen data.

GrowthHacker — Mitacs Globalink Research Internship June–August 2025
University of British Columbia Okanagan | Supervisor: Dr Fatemeh Fard
- Developed, integrated and expanded an existing two-agent prototype into the GrowthHacker experimental workflow, extending its use beyond the initial single-notebook proof of concept.
- Applied GrowthHacker experimental workflow to SCOPE-RL notebooks by automatically selecting execution environments and reusing existing models and datasets instead of re-training models and generating data each time during optimisation.
- Discovered an inconsistently applied scoring methodology between baseline and candidate iterations and rectified by using a unified scoring method to allow for fair comparison between iterations.
- Continued experiments and manuscript revisions beyond the internship, contributing as a co-first author to *GrowthHacker*, accepted by ACM TOSEM.

Meer Labs Limited June - September 2024
Artificial Intelligence Researcher
- Developed multi-agent workflows for quantitative finance research, progressing from a CrewAI prototype to a framework-independent implementation.
- Implemented agents for data collection, analysis and report generation using Python.

SELECTED ACADEMIC PROJECTS

Geometric Deep Learning for Quantum Financial Market Dynamics

Undergraduate dissertation | Birmingham City University | May 2026

- Independently developed and evaluated a financial forecasting model combining hyperspherical embeddings, quantum-probabilistic regime representations and GATv2 graph attention, under Dr Edlira Vakaj's supervision.
- Built a 44-million-record financial dataset retaining delisted equities to support multi-asset forecasting experiments.
- Designed walk-forward evaluation and ablation studies across ten model configurations and 22 folds, with a 20-trading-day purge interval.
- Analysed component contributions and the respective trade-offs across return-ranking correlation, directional accuracy and portfolio risk; found that the geometric-plus-quantum configuration achieved higher return-ranking correlation than the full architecture.

Hate Speech Detection Using Traditional and Transformer-Based Text Classification

Natural Language Processing coursework | Birmingham City University | May 2026

- Implemented and compared nine classification setups consisting of bag-of-words and TF-IDF baselines, frozen DistilBERT embeddings, end-to-end DistilBERT fine-tuning, and zero-shot BART.
- Created a preprocessing pipeline for the task, including retaining of task-relevant pronouns that describe targeting during stop-word removal. Class weighting was employed for logistic regression and SVM classifiers to combat the imbalanced classes in the training set.
- Models were evaluated based on macro F1 scores, as well as hate class precision, recall, and training and inference times. The tradeoffs between these metrics, specifically the risk of not capturing all hate speech versus generating false positives, were analysed in detail.
- Analysed the limited performance advantage of fine-tuning over traditional classifiers and the high-recall, low-precision behaviour of zero-shot classification approach.

MS-GeoGNN — Geometric Deep Learning for 3D Semantic Segmentation

Deep Neural Networks and Ethics coursework | Birmingham City University | September–December 2025

- Constructed a multi-scale geometric graph neural network for 19-class semantic segmentation of 3D point clouds reconstructed from stereo-disparity maps in the Cityscapes dataset.
- Modified the architecture's batch-aware sparse k-nearest-neighbour attention to allow use of scenes with 40,960 points on an 80 GB Nvidia H100 GPU.
- Recorded 71.1% overall accuracy and 16.92% mean intersection over union on the project validation data, with near-zero performance for safety-critical classes including person, rider and bicycle.
- Explored possible explanations for poor safety-critical class performance despite high overall accuracy, including sampling erasure as a potential cause of the 'background trap'.

TEACHING EXPERIENCE

Birmingham City University February 2025 - June 2026

Visiting Demonstrator

• Supported computer science practicals and helped students with programming exercises and AI concepts.

ACADEMIC PRESENTATIONS

Undergraduate dissertation poster defence May 2026

Geometric Deep Learning for Quantum Financial Market Dynamics

• Presented and defended undergraduate dissertation research at Birmingham City University.

TECHNICAL SKILLS

Programming: Python, Java, JavaScript, SQL

Machine learning: PyTorch, TensorFlow, PyTorch Geometric, scikit-learn, Hugging Face Transformers, MONAI

Agent frameworks: CrewAI, LangChain, LlamaIndex, LLM APIs

AI development tools: Cursor, Codex, Claude Code, Hermes Agent

MLOps and deployment: MLflow, Apache Airflow, Docker, Kubernetes, FastAPI

Databases: PostgreSQL, MongoDB

Research tools: Git, Jupyter, LaTeX, GPU computing

AWARDS AND TRAINING

Mitacs Globalink Research Internship | UBC Okanagan | 2025

Enterprise Design Thinking Practitioner (IBM) | 2024

Cyber Security Practices Level 3 | 2023

Foundations of Project Management (Google) | 2022

ADDITIONAL PROFESSIONAL EXPERIENCE

Self-employed audio engineer and music producer June 2016 - December 2022

• Engineered audio for film productions and led recording sessions for music labels.

• Managed recording and production projects with technical requirements and deadlines.